UNIVERSITY OF RWANDA

College of Science and Technology

Department of Computer and Software Engineering

MOBILE COMMUNICATIONS

Unit 3 Mini Projects

LABORATORY REPORT

Frequency-reuse planning, handover decision logic, cellular capacity planning, and finite-state-machine connection management

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Module Mobile Communications — Unit 3 Mini Projects

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GIT REPOSITORY LINK: **https://github.com/Blandon2003/MC_MP_Unit3_Lab.git**

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Figures and tables are numbered by the mini-project to which they belong. The complete Python implementation, CSV logs and JSON result files accompany this report.

1. Introduction

The three mini-projects in Unit 3 move from static radio planning to dynamic mobility management. Mini Project 3.1 builds the geometric foundation: hexagonal clusters, the co-channel reuse ratio Q, and the handover decision rules that keep a moving device attached to the best cell. Mini Project 3.2 turns the same geometry into a capacity-planning tool and asks how an operator should divide a fixed spectrum allocation between cells, and how cell splitting and sectoring create capacity when the reuse factor is already small. Mini Project 3.3 formalises the connection lifecycle as a finite-state machine, including random access and radio-link failure, and compares two handover policies on a scripted three-cell drive.

All experiments were implemented in Python 3.12 with NumPy for numerical work and Matplotlib for figures. Random seeds are fixed, and every numeric result shown in this report is written to CSV/JSON by the scripts listed in Appendix A. The discussion therefore reports what the simulations actually produced rather than expected values, and each paragraph refers to the specific figure or table generated by the code.

1.1 Tools and reproducibility

- Simulation: Python 3.12, NumPy 1.26.4 (arrays, random walks, noise).
- Visualisation: Matplotlib 3.10.0 (hexagon layouts, time series, sweep heatmap, state diagram).
- Reporting: python-docx 1.2.0; the same scripts regenerate every figure and table.
- Seeds: every random process uses a fixed seed, so the reported counts are exactly reproducible.
- No network emulator or external data set is used; all signals are synthetic, as permitted by the lab manual for Mini Projects 3.1 and 3.3.

The report is organised as three linked studies. Section 2 covers frequency-reuse geometry and handover decisions; Section 3 converts the same geometry into capacity planning; Section 4 models the connection lifecycle and policy-driven handover as a finite-state machine. Section 5 draws the three threads together, and the appendices list the submitted files and reproduce the core code.

2. Mini Project 3.1 — Frequency-Reuse Planning and Handover Decision Logic

2.1 Objective

The objective is twofold: (a) build a hexagonal frequency-reuse generator that computes the co-channel reuse distance for different cluster sizes N, and (b) build a handover-decision simulator that applies a hysteresis margin and a time-to-trigger (TTT) to synthetic serving and neighbour RSRP readings, so that the ping-pong effect and its suppression can be observed on the same signal trace.

2.2 Part (a): Hexagonal reuse planning

The layout uses pointy-top hexagons in axial coordinates (q, r). A cell centre is placed at $x = \sqrt{3}\cdot R\cdot(q + r/2)$, $y = 1.5\cdot R\cdot r$, and all cells within a chosen number of rings are enumerated using $\max(|q|, |r|, |q+r|)$. A hexagonal cluster size must satisfy $N = i^2 + ij + j^2$. The pair (i, j) defines two tiling vectors, u = (i, j) and v = (–j, i+j), whose integer combinations generate the cluster lattice. Frequency groups are assigned by taking the cosets of that lattice: two cells share a group exactly when their axial difference is an integer combination of u and v. The code finds the nearest cell of each coset and numbers the N distinct cosets 1..N. This guarantees that co-channel cells repeat at the theoretical distance $D = \sqrt{(3N)}\cdot R$, which a naive modulo formula can violate; for N = 4, for example, a simple linear label would colour adjacent cells with the same group.

Table 3.1 — Cluster parameters, reuse ratio and layout coverage.

N	i	j	Q = √(3N)	Groups visible	Cells drawn
1	1	0	1.732	1	91
3	1	1	3.000	3	127
4	2	0	3.464	4	169
7	2	1	4.583	7	217
9	3	0	5.196	9	271
12	2	2	6.000	12	331

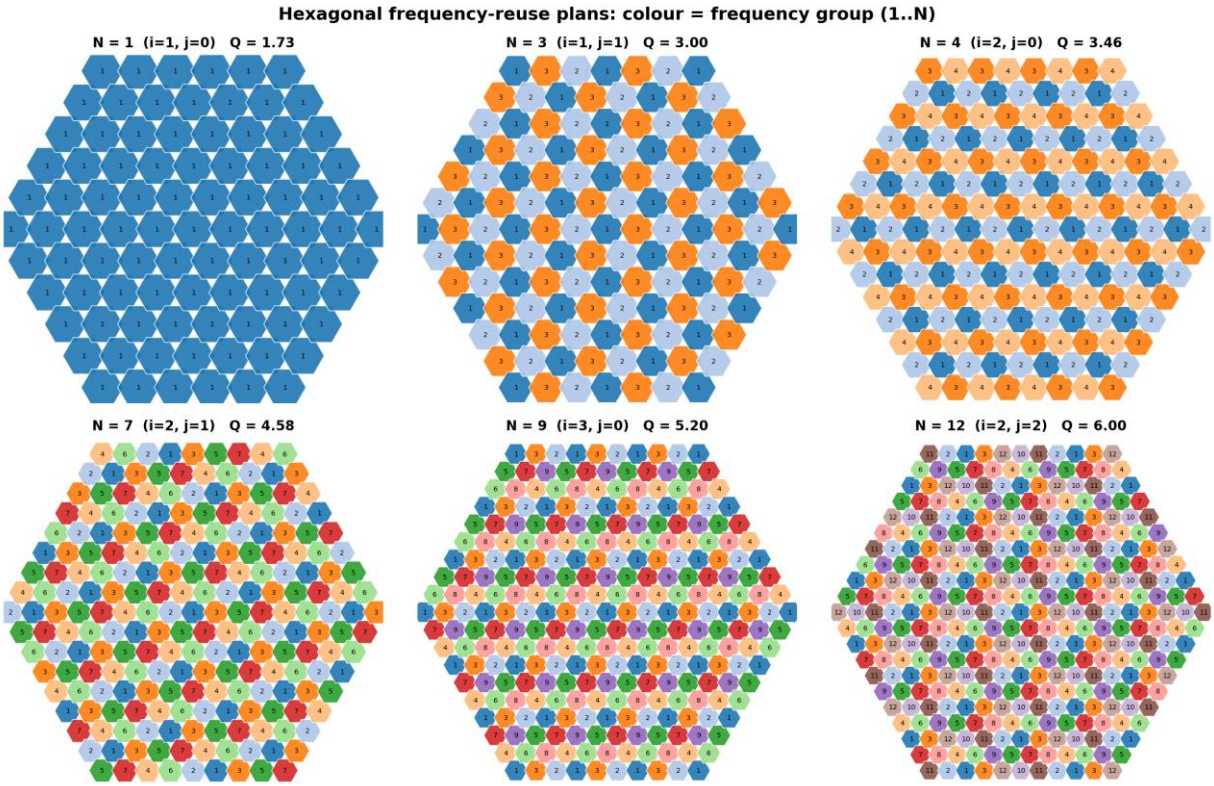


Figure 3.1 — Hexagonal reuse layouts for N = 1, 3, 4 and 7 (required) and N = 9, 12 (extension). Colour identifies the frequency group; every same-coloured hexagon is a co-channel cell at distance D = Q·R.

The six panels confirm the expected geometry. With N = 1 every cell uses the same frequency, so capacity is maximal but no co-channel separation exists. N = 3 and N = 4 introduce the familiar three- and four-colour tilings, with Q = 3.00R and 3.46R. N = 7, the classic cellular planning value, places the nearest co-channel cell at 4.58R and produces a visually clean pattern with well-separated same-numbered cells. The extensions, N = 9 and N = 12, increase the separation to 5.20R and 6.00R at the cost of larger clusters.

2.3 Part (b): Handover decision simulation

Two 300-sample RSRP traces were generated for a serving cell A and a neighbour B. Cell A has a gentle decline and then a steep drop after sample 190; cell B rises and then flattens, so the curves overlap for roughly 90 samples before A deteriorates. Both traces contain a cumulative Gaussian random walk (shadowing) and independent Gaussian measurement noise (σ = 0.8 dB). Three policies were run on exactly the same sample: naive (hand over on any positive margin), hysteresis only (3 dB), and hysteresis plus TTT (3 dB held for five consecutive samples). After every handover the serving and neighbour roles swap, so a return handover is possible and is counted. A handover is classified as ping-pong when the device returns to a cell it left fewer than 30 samples earlier.

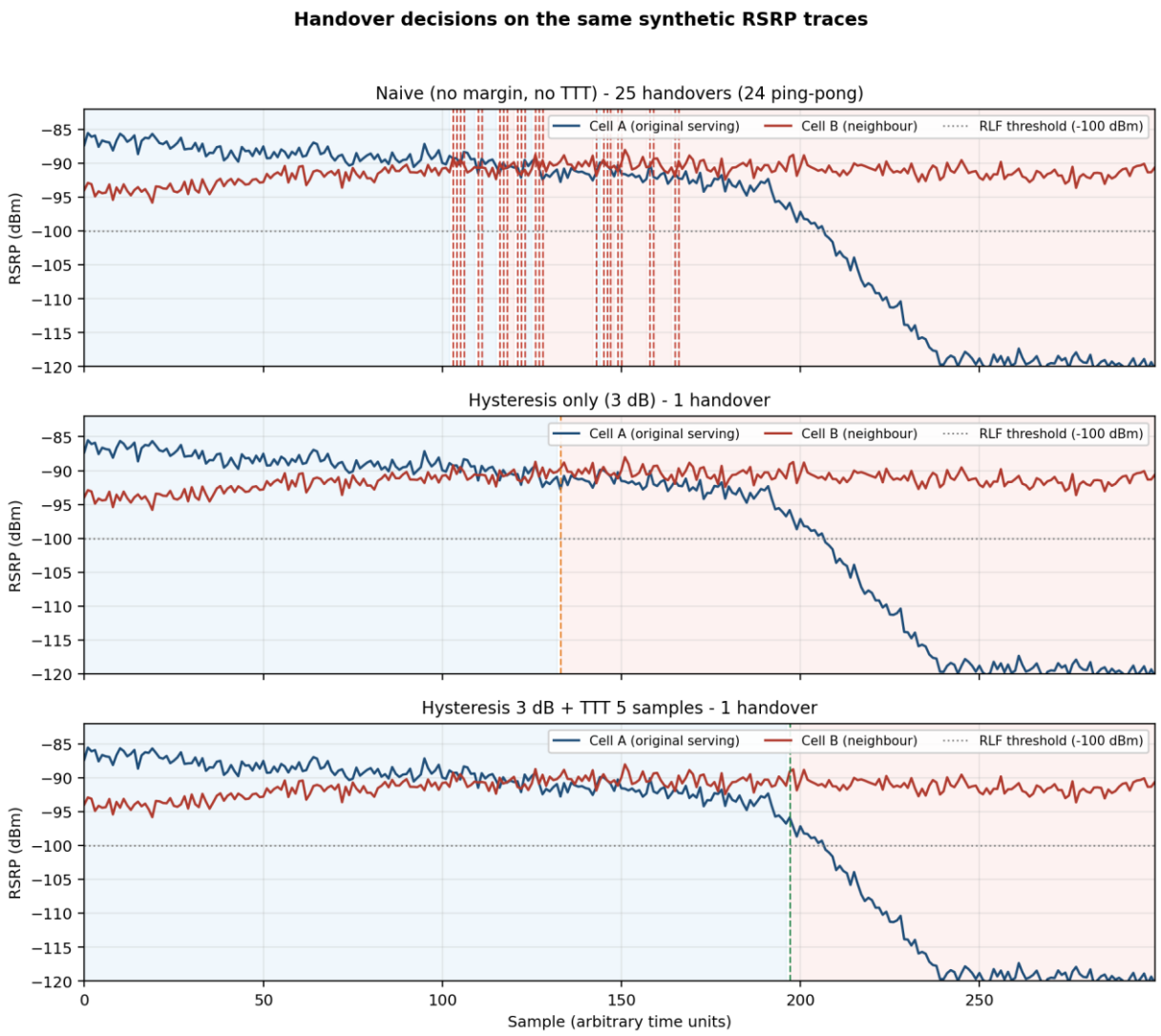


Figure 3.2 — The same serving/neighbour traces under the three policies. Shading indicates the active serving cell, dashed vertical lines mark handovers, and the dotted line is the −100 dBm planning/RLF threshold.

Table 3.2 — Handover-count comparison over the 300-sample trace.

Configuration	Handovers	Ping-pong	Min. serving RSRP (dBm)	First handover (sample)
Naive (no margin, no TTT)	25	24	-93.62	103
Hysteresis only (3 dB)	1	0	-93.62	133
Hysteresis 3 dB + TTT 5 samples	1	0	-96.75	197

2.4 Interpretation of Q and the handover counts

Table 3.1 shows Q rising as √(3N): 1.73 for N = 1, 3.00 for N = 3, 3.46 for N = 4, 4.58 for N = 7, 5.20 for N = 9 and 6.00 for N = 12. Smaller clusters therefore place same-frequency cells closer together, which raises spatial reuse but lowers the carrier-to-interference ratio. In Figure 3.1 the N = 1 layout uses one group everywhere, N = 3 and N = 4 repeat groups only a few cell widths away, and N = 7 shows clearly separated same-numbered cells with a Q 32% larger than N = 4. The handover traces show the other half of the trade-off. Without a margin or timer the decision follows every noise-driven crossover: 25 handovers, 24 of them returning to a cell left less than 30 samples earlier, which is the ping-pong effect. A 3 dB hysteresis reduces this to one handover, but it still fires early on a noise spike at sample 133 while the serving signal is −92.8 dBm. Adding a five-sample TTT moves the decision to sample 197, after the serving cell has genuinely deteriorated, and

removes the remaining oscillation; the cost is a 3 dB weaker serving signal at decision time (−96.8 versus −93.6 dBm). Hysteresis therefore suppresses ping-pong and TTT suppresses short-lived false triggers, but an excessive TTT causes a too-late handover: a 3 dB margin with a 20-sample TTT hands over at −103.6 dBm, below the −100 dBm planning floor used here.

2.5 Extended lab tasks

2.5.1 Larger reuse factors: N = 9 and N = 12

Extending the generator to N = 9 (i = 3, j = 0) and N = 12 (i = 2, j = 2) continues the same trend: Q rises to 5.20 and 6.00, so co-channel cells move farther apart, but the cluster consumes more of the spectrum and only 27 or 20 of the 250 channels remain in each cell (Table 4.2). The SIR estimate improves from 18.7 dB at N = 7 to 20.9 and 23.3 dB, a gain of only 2.2 and 4.6 dB for capacity losses of 23% and 43% relative to N = 7. Beyond N = 7 the capacity–interference trade-off is therefore dominated by diminishing SIR returns; larger clusters are justified only where coverage reliability, not capacity, is the binding constraint.

2.5.2 Strongest-of-three selection

A third neighbour (C) was generated with its own trend, random walk and noise, and the triggers were changed to select the strongest of the available cells at each decision point. The naive policy produced 26 handovers (24 ping-pong), while both tuned configurations produced a single handover and no ping-pong. Selecting the strongest neighbour prevents a handover to a weaker cell, but it does not by itself remove oscillation between the two strongest cells; the margin and TTT are still required. The result also shows that the handover count is sensitive to the number of candidates, which is why mobility robustness optimisation treats neighbour lists and thresholds together.

2.5.3 Sweeping margin and TTT

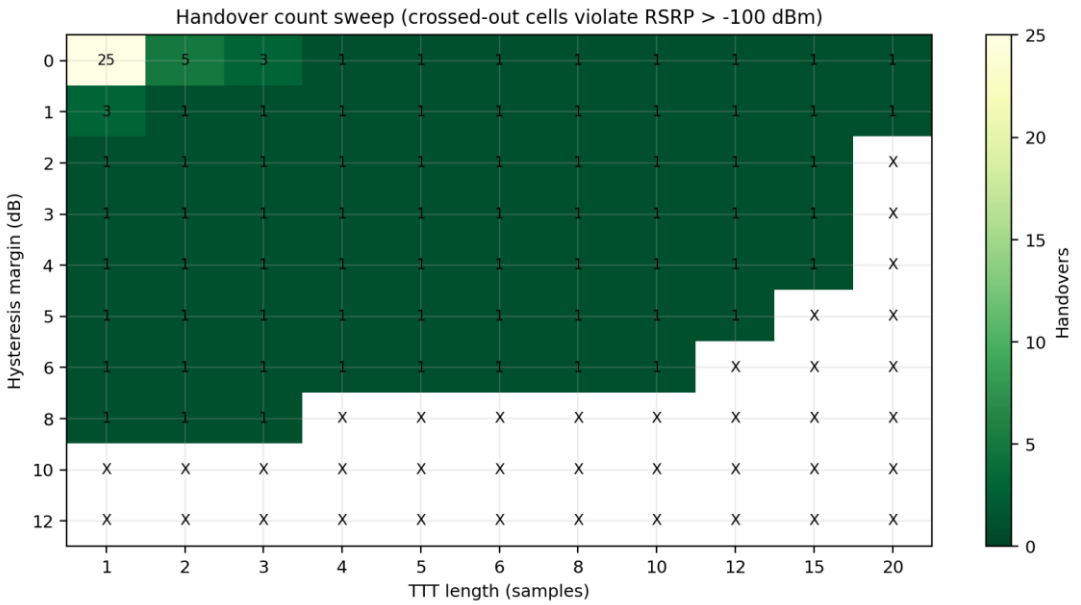


Figure 3.3 — Handover count over a 10 × 11 margin/TTT grid. Cells marked X violate the −100 dBm serving-RSRP floor before the handover completes.

Table 3.3 — Selected points from the margin/TTT sweep.

Margin (dB)	TTT (samples)	Handovers	Min. serving RSRP (dBm)	Meets −100 dBm floor
0	1	25	-93.62	yes
3	5	1	-96.75	yes
3	20	1	-103.61	no

Margin (dB)	TTT (samples)	Handovers	Min. serving RSRP (dBm)	Meets −100 dBm floor
5	20	1	-103.74	no
8	20	1	-110.21	no
12	20	1	-113.87	no

Every configuration with a margin of at least 1 dB and a TTT of at least two samples reduces the count to one, because the trace contains a single genuine cell transition. The sweep is nevertheless useful for exposing the other failure mode. With a 3 dB margin and a 20-sample TTT the handover occurs at sample 212, by which time the serving cell has fallen to −103.6 dBm; larger margins delay it further, reaching −113.9 dBm at 12 dB/20 samples. Those points are infeasible even though their handover count is the minimum possible. The sweep therefore selects 3 dB and five samples as a robust operating point: it yields one handover, keeps the minimum serving RSRP at −96.8 dBm, and leaves roughly 3 dB of headroom above the floor. This is a small-scale version of Mobility Robustness Optimisation: minimise unnecessary handovers subject to a coverage/reliability constraint.

2.5.4 Faster-moving device

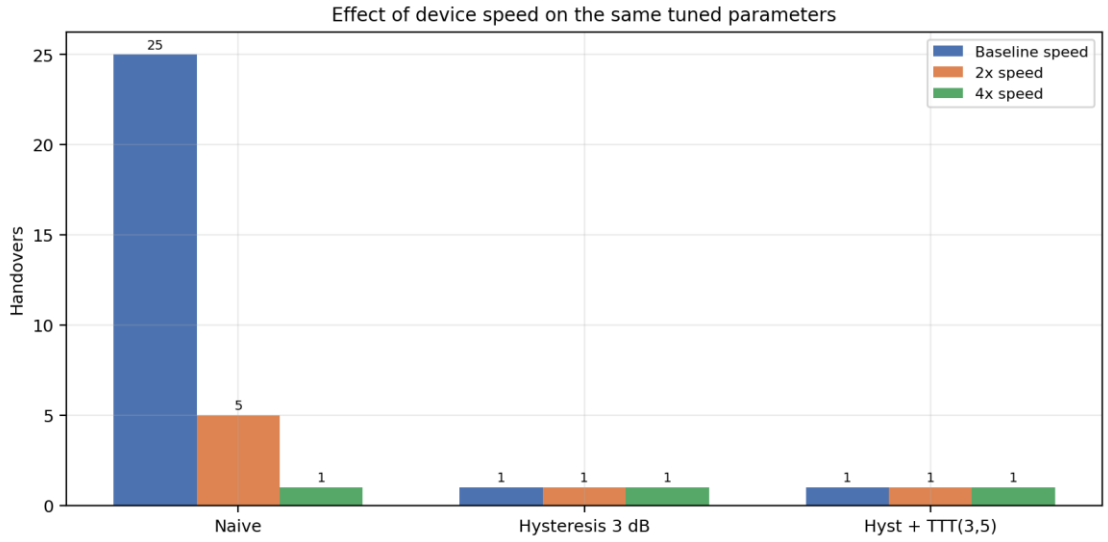


Figure 3.4 — Handover counts when the trend slope and random-walk variance are scaled to model a faster device.

Table 3.4 — Handovers and minimum serving RSRP versus device speed (count and dBm).

Speed	Naive	Hysteresis 3 dB	Hysteresis + TTT(3,5)
Baseline speed	25 (-93.6 dBm)	1 (-93.6 dBm)	1 (-96.8 dBm)
2x speed	5 (-91.4 dBm)	1 (-92.3 dBm)	1 (-93.3 dBm)
4x speed	1 (-90.5 dBm)	1 (-92.1 dBm)	1 (-93.8 dBm)

Increasing the trend slope and random-walk variance makes the device traverse each overlap region in fewer samples. The naive policy consequently drops from 25 to 5 and then 1 handover: there is simply less time for noise to produce repeated crossovers. The tuned policy remains at one handover in all three cases, but its minimum serving RSRP falls from −96.8 to −93.4 and −93.8 dBm because the decision is reached earlier in absolute signal terms. A fixed sample-count TTT also represents a shorter time and a shorter distance at higher speed, so the protection against a fast fade weakens. At highway or high-speed-rail velocities the TTT should be specified in milliseconds (for example 160–320 ms) and evaluated against the expected fade duration, rather than fixed in samples.

2.5.5 Asymmetric offset for load balancing

A 2 dB offset was applied in favour of neighbour B, so the effective entering threshold became 1 dB instead of 3 dB. The handover count stayed at one, but the decision moved from sample 197 to sample 170, 27 samples earlier. The biased threshold transfers load to the favoured cell earlier and keeps the device on the preferred cell longer, but it also increases the risk of handing a marginal user to a cell that is not actually better and of unbalancing the two cells. Load-balancing offsets must therefore be bounded and combined with a utilisation check, not applied as a permanent signal-quality bias.

2.5.6 Sectoring and the interference geometry

Replacing each omni cell with three 120° sectors changes which neighbours can interfere on a given frequency group. In the omni plan all six first-tier co-channel cells radiate into the cell, whereas a 120° sector receives mainly from the two co-channel cells that fall inside its main beam. With a path-loss exponent of four, the SIR estimate improves by a factor of three (about 4.8 dB). This is why a three-sector site at N = 4 (18.6 dB in the estimate of Section 3) can match an omni site at N = 7 (18.7 dB) while carrying 1.77× the traffic. Sectoring also turns every cell into three smaller coverage areas, which adds inter-sector handovers and antenna/backhaul cost, but it avoids the new sites that cell splitting requires.

3. Mini Project 3.2 — Cellular Capacity and Frequency-Reuse Planning

3.1 Objective and inputs

The objective is a planning tool that computes cellular system capacity under different reuse factors and uses it to analyse the coverage-versus-capacity trade-off, including cell splitting and sectoring as alternative capacity levers.

Table 4.1 — Planning inputs used by the capacity tool.

Parameter	Symbol	Value
Total spectrum allocation	B	50 MHz
Channel bandwidth	Bc	200 kHz
Coverage area	A	400 km²
Baseline cell radius	R	2.0 km
Candidate reuse factors	N	1, 3, 4, 7, 9, 12
Path-loss exponent (SIR only)	n	4
Sectors per site (sectoring case)	—	3

3.2 Method

With total spectrum B = 50 MHz and channel bandwidth Bc = 200 kHz, the operator has $C_{total} = \lfloor B/Bc \rfloor = 250$ traffic channels. For a fixed coverage area A = 400 km² and cell radius R = 2 km, the number of hexagonal cells is $M = \lfloor A / (2.598 \cdot R^2) \rfloor = 38$, where $2.598 \cdot R^2$ is the area of one hexagon ($3\sqrt{3}/2 \cdot R^2$). For each reuse factor, channels per cell are $\lfloor C_{total}/N \rfloor$ and total system capacity is that value times M. The reuse ratio is $Q = \sqrt{(3N)}$. To make the interference trade-off concrete, the co-channel signal-to-interference ratio is approximated as $SIR = Q^n/6$ for an omni site with six first-tier interferers, with path-loss exponent n = 4. These are planning estimates: they ignore shadowing, antenna patterns and trunking, but they are sufficiently accurate to compare candidate reuse factors.

The implementation keeps every input in a single configuration block at the top of the script, computes the derived quantities in a pure function, and writes the table to CSV before plotting. The cell count is floored because a partial hexagonal cell cannot carry a full traffic load in this simplified model.

3.3 Capacity results

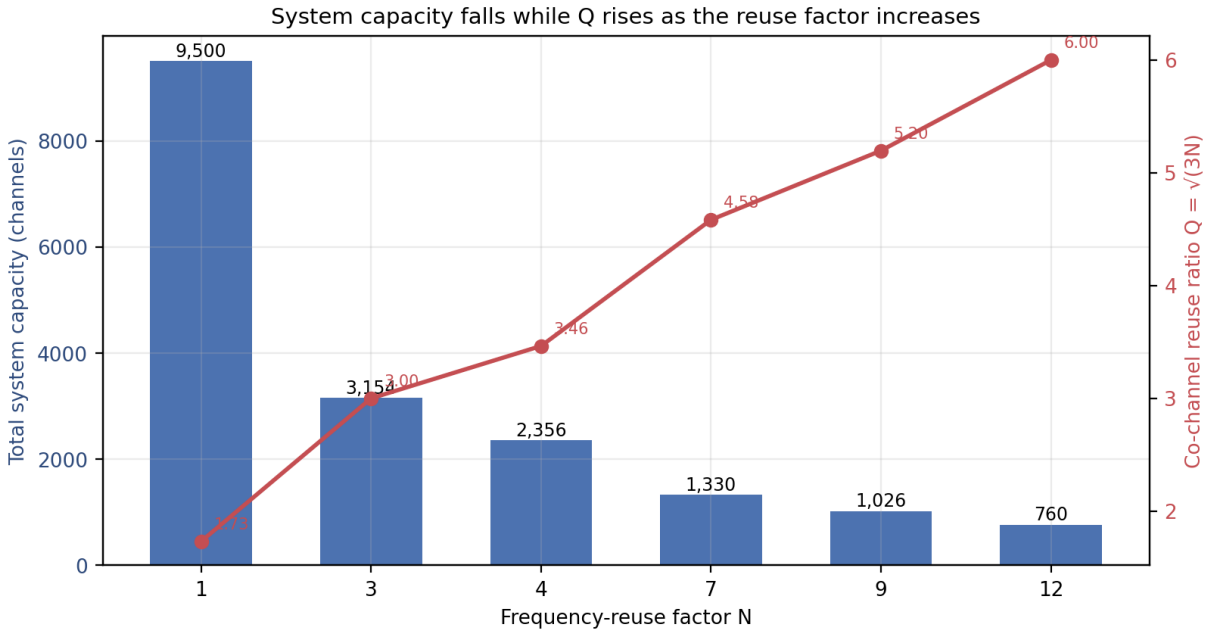


Figure 4.1 — Total system capacity (bars, left axis) and reuse ratio Q (line, right axis) for the six candidate reuse factors.

Table 4.2 — Channels, capacity and co-channel SIR for each candidate reuse factor.

N	Q	Channels / cell	Cells	System capacity	SIR estimate (dB)
1	1.732	250	38	9,500	1.8
3	3.000	83	38	3,154	11.3
4	3.464	62	38	2,356	13.8
7	4.583	35	38	1,330	18.7
9	5.196	27	38	1,026	20.9
12	6.000	20	38	760	23.3

The table exposes the central planning conflict. Capacity falls almost as 1/N because each cell receives fewer channels, while Q and the SIR estimate rise as √N. The practical question is therefore not which row has the largest number, but where the SIR crosses the minimum acceptable value for the service being offered. N = 1 is a useful upper bound but not a deployable plan: a 1.8 dB SIR would make the cell unusable except at the exact cell centre.

3.4 Cell splitting and sectoring

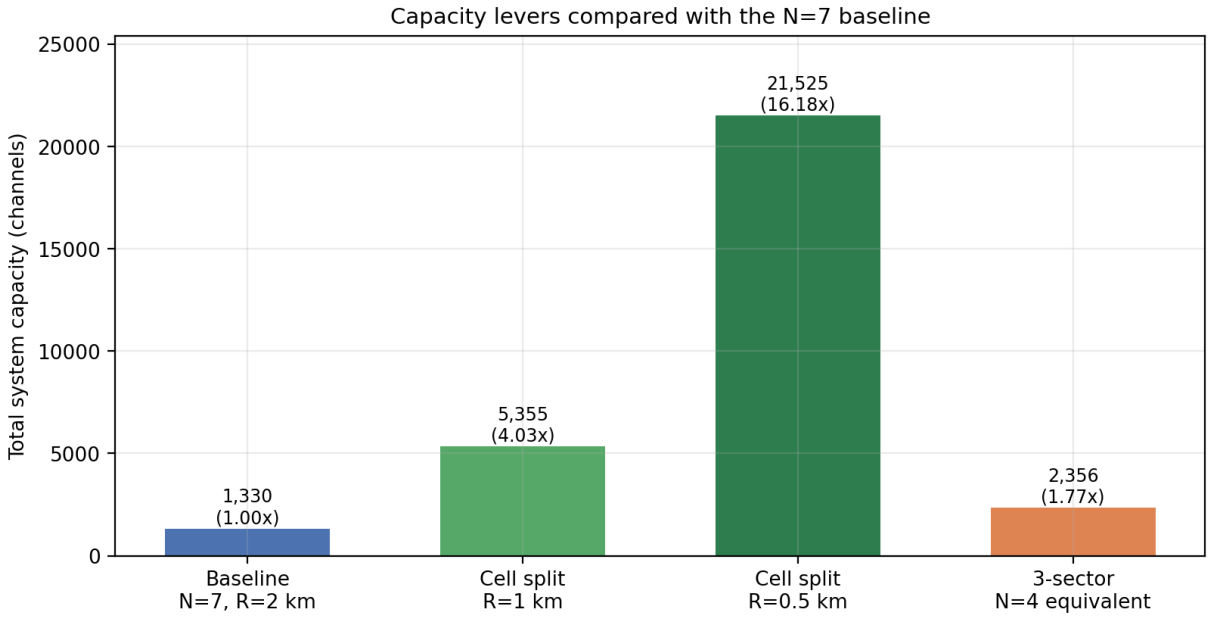


Figure 4.2 — Capacity of the N = 7 baseline compared with halving and quartering the cell radius and with a three-sector N = 4 plan.

Table 4.3 — Quantified capacity levers (cell splitting and sectoring).

Plan	Cell radius	Cells	Capacity (channels)	Gain vs baseline
Baseline N = 7	2 km	38	1,330	1.00×
Cell split N = 7	1 km	153	5,355	4.03×
Cell split N = 7	0.5 km	615	21,525	16.18×
3-sector, N = 4	2 km	38	2,356	1.77×

Halving R adds four times as many cells and increases capacity from 1,330 to 5,355 channels (4.03×); quartering R gives 21,525 channels (16.18×) in the same 400 km². The sectoring row is not a threefold multiplication of the channel count: the same 250 channels are divided among the sectors, and the gain comes from being able to use N = 4 instead of N = 7 at comparable SIR. The resulting capacity is 2,356 channels (1.77× the N = 7 baseline) with an estimated sector SIR of 18.6 dB, essentially equal to the 18.7 dB of the omni N = 7 site.

3.5 Analysis: choosing a reuse factor

With 50 MHz and 200 kHz channels the operator has 250 traffic channels. For a fixed 400 km² area and R = 2 km, 38 hexagonal cells fit, so total capacity is 38 × ⌊250/N⌋. Table 4.2 shows the complete trade-off. N = 1 offers 9,500 channels but Q = 1.73, and the six-interferer estimate gives only 1.8 dB SIR, far below the 10–15 dB usually needed for acceptable speech and data, so it is not deployable. N = 3 (3,154 channels, 11.3 dB) and N = 4 (2,356 channels, 13.8 dB) are plausible dense-urban choices when sectoring or interference coordination controls the additional interference. N = 7 (1,330 channels, 18.7 dB) sacrifices 44% of the N = 4 capacity for 4.9 dB more SIR; N = 9 and N = 12 raise SIR to 20.9 and 23.3 dB but reduce capacity to 1,026 and 760 channels. The SIR gain per added reuse step clearly diminishes, while the capacity loss per step does not.

In a dense urban deployment the traffic demand per square kilometre is high, sites are difficult to acquire, and users are close to many cells. The operator should therefore keep N small (3–4), use three-sector sites and possibly a second carrier, and rely on interference coordination and handover rather than large clusters. In a rural deployment the demand per square kilometre is low and the network is coverage-limited, so a larger

N (7 or 9) is preferable: the extra SIR improves edge reliability, and the capacity loss does not matter when the sites are lightly loaded.

Once N is already small, reducing it further is the wrong lever: N = 1 is interference-limited, N = 2 does not tile a hexagonal grid, and N = 3 leaves very little SIR margin. Cell splitting creates capacity by increasing the number of cells. Halving R quadruples the cell count and raises capacity from 1,330 to 5,355 channels (4.03×); quartering R gives 21,525 channels (16.18×). The gain is real but requires new sites, backhaul and antennas, and in a dense city the required sites may simply not exist. Sectoring is cheaper: three 120° sectors reduce the first-tier interferers seen in any direction from six to about two, a theoretical 4.8 dB SIR gain. That allows N = 4 to reach 18.6 dB, matching the 18.7 dB of an omni N = 7 site, while capacity rises from 1,330 to 2,356 channels (1.77×) without new sites. This is why splitting and sectoring are preferred once N is small: they add capacity through spatial reuse and interference control, whereas shrinking N still further only raises interference. Sectoring adds inter-sector handovers and antenna cost, and these figures ignore trunking efficiency; an Erlang-B calculation with a target blocking probability would be the natural next step.

4. Mini Project 3.3 — Finite-State Machine for Connection and Handover

4.1 Objective

The objective is to model the device's connection lifecycle as an explicit finite-state machine — Idle, Random-Access, Connected and Handover-in-progress, plus a Radio-Link-Failure path — and to drive it through a scripted three-cell (A→B→C) mobility scenario under two configurable handover policies.

4.2 State machine design

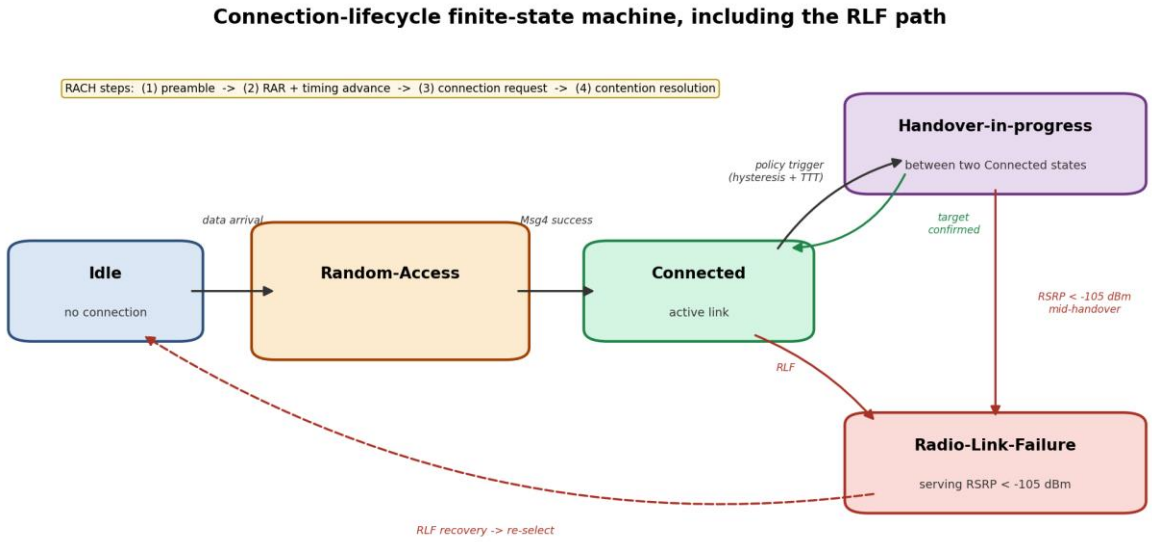


Figure 5.1 — Connection-lifecycle FSM. The RACH sub-steps are shown at the top; every transition into Handover-in-progress is produced by the selected policy.

Table 5.1 — Complete transition table, including the RACH and RLF paths.

From	To	Trigger / condition	Action
Idle	Random-Access	Connection request or paging; best cell above RLF threshold	Select cell, send preamble (Msg1)
Random-Access	Random-Access	Random Access Response received	Apply timing advance (Msg2)
Random-Access	Random-Access	Grant received	Send RRC connection request (Msg3)
Random-Access	Connected	Contention resolution succeeds	Attach to cell (Msg4)
Connected	Handover-in-progress	Naive: any positive margin. Tuned: neighbour > serving + 3 dB for 5 samples	Select strongest neighbour; begin HO
Handover-in-progress	Connected	Target access confirmed	RRC reconfiguration complete
Connected	Radio-Link-Failure	Serving RSRP < −105 dBm	Declare RLF
Handover-in-progress	Radio-Link-Failure	Serving RSRP < −105 dBm during HO	Declare RLF (too-late HO)
Radio-Link-Failure	Idle	RLF recovery	Release connection; re-select
Idle (after RLF)	Random-Access	Cell above threshold	Re-access through RACH

4.3 Random-access representation

RACH is represented as the four logical message steps from the unit notes. At t = 0 the device selects the strongest cell and transmits the preamble (Msg1). The network answers with a Random Access Response carrying the timing advance, after which the device sends the RRC connection request (Msg3) and waits for contention resolution (Msg4). The FSM records each step as a timestamped entry in the transition log and only enters Connected after Msg4 succeeds, so the Idle→Connected transition is explicitly RACH-driven rather than assumed. The same four-step sequence is used for re-access after an RLF.

4.4 Scripted scenario and policies

The device travels from the centre of cell A, through B, to C. Cell centres are placed at 0, 2 and 4 km, and 400 samples cover the 3.9 km route, so the A/B and B/C crossings occur at about samples 100 and 300. Each cell's RSRP follows a log-distance path-loss model (−60 − 30 log10(d/0.1) dBm), with a cumulative Gaussian shadowing term (σ_step = 0.25 dB) and independent measurement noise (σ = 0.8 dB). This produces genuine overlap regions in which the two strongest cells are within a few dB, as required for a real handover decision. Both policies consume the same sample and differ only in the decision rule: the naive policy hands over as soon as any neighbour is stronger, while the tuned policy requires a 3 dB margin held for five consecutive samples. The transition log records time, source state, destination state, cell and trigger for every change.

4.5 Results

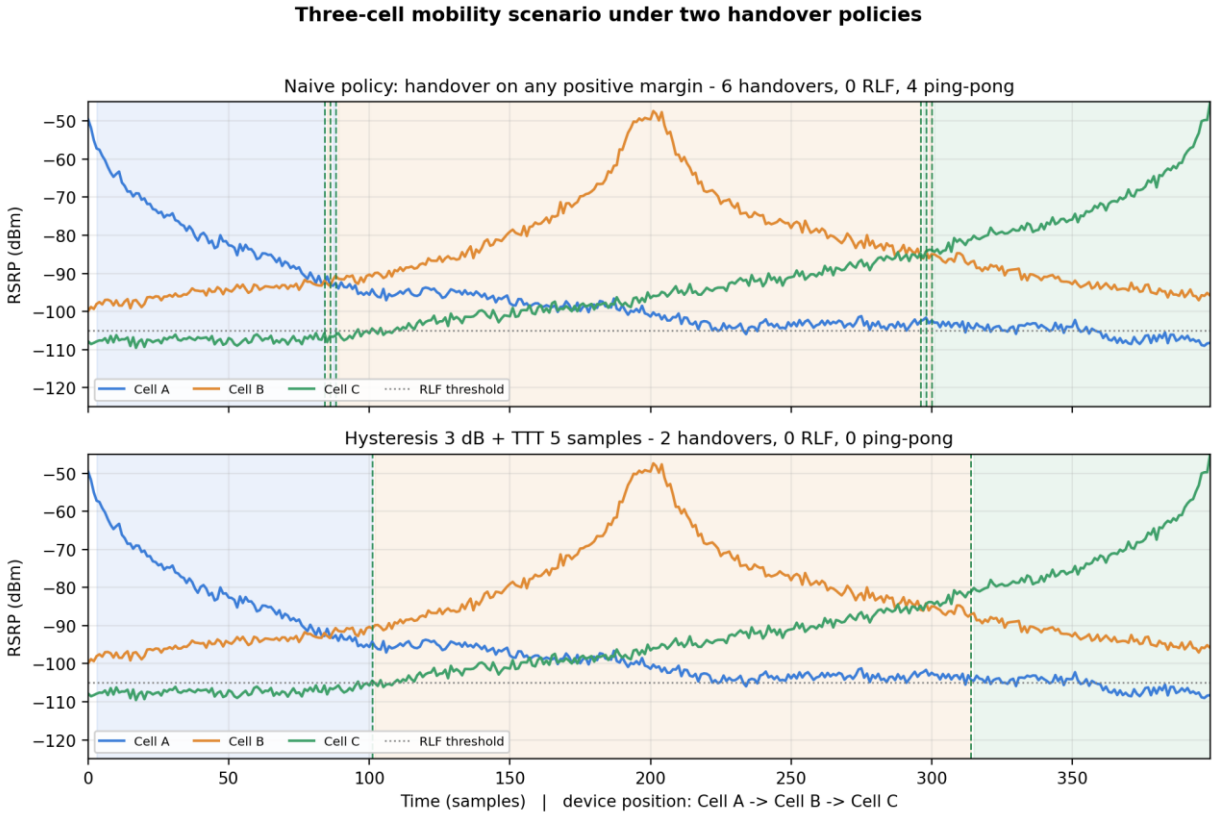


Figure 5.2 — The A→B→C RSRP scenario under the naive and tuned policies. Background colour shows the serving cell, green dashed lines are handovers and the dotted line is the RLF threshold.

Table 5.2 — Handover, RLF and ping-pong counts per policy (30-sample ping-pong window).

Policy / case	Handovers	RLF	Ping-pong	Mean serving RSRP (dBm)	Min. serving RSRP (dBm)
Naive, nominal	6	0	4	-76.24	-93.27
Hysteresis + TTT, nominal	2	0	0	-76.45	-95.62

Policy / case	Handovers	RLF	Ping-pong	Mean serving RSRP (dBm)	Min. serving RSRP (dBm)
Naive, deep fade	8	0	5	-76.70	-93.27
Hysteresis + TTT, deep fade	3	1	0	-76.89	-100.02

Table 5.3 — Excerpt from the tuned-policy transition log (full CSV logs are submitted).

Time	State transition	Cell	Trigger
0	Idle → Random-Access	A	Connection request; RACH preamble (Msg1)
1	Random-Access	A	Random Access Response + timing advance (Msg2)
2	Random-Access	A	RRC connection request (Msg3)
3	Random-Access → Connected	A	Contention resolution success (Msg4)
101	Connected → Handover-in-progress	B	B exceeds A by >3 dB for 5 samples
102	Handover-in-progress → Connected	B	Target B access confirmed
244	Connected → Radio-Link-Failure	B	Serving RSRP −105.7 < −105 dBm (fade case)
245	Radio-Link-Failure → Idle	B	RLF recovery; release and re-select
246	Idle → Random-Access	C	Re-access; RACH preamble (Msg1)
249	Random-Access → Connected	C	Contention resolution success (Msg4)
314	Connected → Handover-in-progress	C	C exceeds B by >3 dB for 5 samples
315	Handover-in-progress → Connected	C	Target C access confirmed

The nominal scenario isolates the policy difference cleanly. The naive policy performs six handovers, four of which are returns within 30 samples (A→B→A→B at samples 84–88 and C→B→C at 296–300); the tuned policy performs only the two intended transitions, A→B at 101 and B→C at 314, with no ping-pong. The tuned policy's minimum serving RSRP is 2.4 dB lower, which is the expected cost of waiting five samples before committing.

4.6 Radio-link-failure stress test

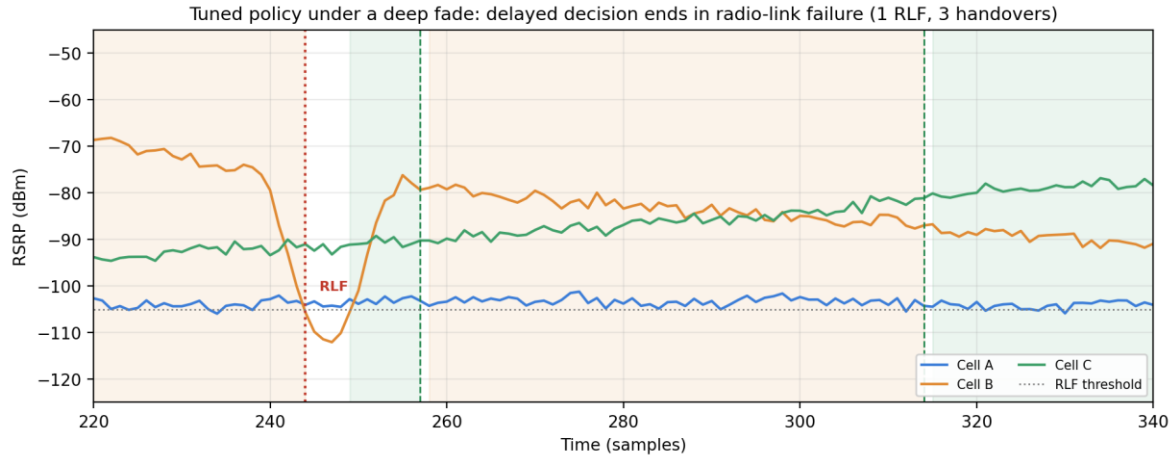


Figure 5.3 — The tuned policy under a 36 dB fade on cell B. B falls below the RLF threshold at sample 244 while the TTT is still running; the connection returns to Idle and re-accesses on C.

A deep fade of up to 36 dB was applied to cell B between samples 238 and 256. The naive policy reacts immediately when C becomes stronger and moves to C at sample 242, before B reaches the RLF threshold. The tuned policy continues to wait for five consecutive samples; B drops to −105.7 dBm at sample 244, the RLF condition fires, and the FSM follows Connected → Radio-Link-Failure → Idle → Random-Access →

Connected(C). This is a concrete demonstration of a too-late handover: the hysteresis and TTT that suppress ping-pong also delay the legitimate handover when the serving link collapses quickly. The RLF count is therefore not an implementation defect but the intended failure path; a production policy would add an event-triggered (A3/A5) report and a shorter TTT for degrading links.

4.7 Comparison of the two policies

Both policies use the same A→B→C traces but different decision logic. The naive policy hands over whenever a neighbour is even fractionally stronger. Because the traces are within 1–2 dB of each other for roughly thirty samples around each crossing, measurement noise repeatedly reverses the ranking: six handovers, four of which return to a cell left within the previous 30 samples, which is genuine ping-pong. The tuned policy requires the neighbour to exceed the serving cell by 3 dB for five consecutive samples. It performs exactly the two intended handovers (A→B at sample 101 and B→C at sample 314) with no ping-pong, and its mean serving RSRP is essentially unchanged (−76.45 versus −76.24 dBm). The price is a later decision: its minimum serving RSRP is 2.4 dB lower than the naive policy's (−95.6 versus −93.3 dBm), because a device that waits five samples continues to move away from the serving cell.

The deep-fade stress case exposes the risk of over-tuning. When cell B fades by up to 36 dB around sample 244, the naive policy reacts within one sample and hands over to C at sample 242, so it records no radio-link failure. The tuned policy is still waiting for its five-sample TTT when B falls to −105.7 dBm, below the −105 dBm RLF threshold, so the connection fails, returns to Idle and re-accesses through RACH to C. As the fade clears, B is re-selected at sample 257 and C again at 314. This is exactly the too-late-handover failure that the RLF path exists to catch. The experiment shows that hysteresis and TTT are complementary but must be dimensioned against the fastest expected fade: the margin controls ping-pong, the TTT suppresses short noise excursions, and the RLF threshold caps the waiting time. A practical policy would use a time-based TTT (for example 160–320 ms), a 2–4 dB margin, a release threshold derived from the link budget, and a short blacklist for a failed cell to avoid immediate re-selection thrash.

5. Discussion and Conclusion

The three mini-projects form a coherent chain. Frequency reuse fixes the geometry: Q = √(3N) grows slowly with N, while the number of channels per cell falls as 1/N, so every increase in reuse distance is paid for in capacity. The handover simulations show that the same trade-off appears in time rather than space: a decision that reacts to every measurement noise spike (naive) produces ping-pong, while a decision that waits (hysteresis + TTT) suppresses it but can arrive too late. The capacity tool makes the spatial trade-off quantitative and shows why operators stop reducing N once it is small and turn to cell splitting and sectoring instead: those levers add capacity without further shrinking the cluster. The FSM then places the handover decision inside the wider connection lifecycle, where RACH and radio-link failure determine whether a decision is merely suboptimal or actually disruptive.

The measured results support four conclusions. First, N = 7 is a sound default for the studied geometry: Q = 4.58 and the estimated 18.7 dB SIR are adequate, while N = 3 or 4 is only viable with sectoring or interference coordination. Second, a 3 dB hysteresis alone is not sufficient: in the trace it fired on a noise spike at sample 133, 64 samples before the genuine transition; the five-sample TTT was needed to eliminate that false trigger. Third, over-tuning is a real failure mode: margin/TTT combinations that reduce the handover count to one can push the serving RSRP below the −100 dBm floor (up to −113.9 dBm in the sweep), and the FSM shows the same effect producing an RLF at −105.7 dBm. Fourth, sectoring is the cheapest capacity lever in this planning exercise: it gives 1.77× capacity at the same SIR estimate by allowing N = 4 instead of N = 7, without new sites.

The main limitations are the simplified propagation model (no terrain, antenna patterns or correlated shadowing between cells), the SIR approximation based on six equal-power interferers, the one-sample handover execution, and the use of traffic channels rather than Erlang capacity. The hexagonal layouts are also drawn without wrap-around, so edge cells see fewer neighbours than an infinite plan. These simplifications do not change the direction of any conclusion, but they mean the absolute SIR and capacity numbers should be treated as planning estimates. Natural extensions are an Erlang-B blocking calculation, a wrap-around hexagonal grid, a time-based TTT in milliseconds, sector-level state machines with inter-sector handovers, and a load-aware neighbour selection policy.

6. References

[1] T. S. Rappaport, Wireless Communications: Principles and Practice, 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2002, ch. 3 (frequency reuse, handover, capacity).

[2] 3GPP, TS 36.331 — Evolved Universal Terrestrial Radio Access (E-UTRA): Radio Resource Control (RRC) protocol specification, current release (RACH, measurement reports, handover).

[3] 3GPP, TS 36.133 — Requirements for support of radio resource management, current release (time-to-trigger, radio-link monitoring and failure).

[4] W. C. Y. Lee, Mobile Cellular Telecommunications: Analog and Digital Systems, 2nd ed. New York: McGraw-Hill, 1995 (sectoring and co-channel interference).

[5] Mobile Communications, Unit 3: Lab Manual — Mini Projects 3.1–3.3, prepared September 2026.

Appendix A — Submitted files and reproduction

Table A.1 — Deliverable package.

File	Purpose
mp31_reuse_handover.py	Hexagonal reuse layouts, Q table, synthetic RSRP traces, three handover policies, sweep, speed, bias and three-cell extensions.
mp32_capacity.py	Capacity-planning calculations, cell-splitting scenarios, sectoring estimates and comparison charts.
mp33_fsm.py	FSM implementation, RACH sequence, both policies, A→B→C scenario, RLF stress test and transition logs.
generate_report.py	Builds this DOCX from the saved JSON/CSV outputs and figures.
figures/	All PNG figures used in the report (200 dpi).
results/	CSV tables and JSON result files for every experiment.

Reproduction is a four-command sequence from the project folder. The scripts have no network dependencies and use only NumPy and Matplotlib:

```
python mp31_reuse_handover.py
python mp32_capacity.py
python mp33_fsm.py
python generate_report.py
```

Each simulation writes its figures to figures/ and its numeric results to results/. The handover logs are also written as CSV so every state transition can be audited independently of the plots.

Appendix B — Core implementation excerpts

The excerpts below show the three pieces of logic that are easiest to get wrong: the coset assignment that makes the reuse layout exact, the common trigger used by the handover policies, and the RACH-driven part of the finite-state machine. The full scripts are submitted with this report.

B.1 Exact frequency-group assignment (Mini Project 3.1)

```
def _in_cluster_lattice(dq, dr, i, j, n):
    a = (i + j) * dq + j * dr
    b = -j * dq + i * dr
    return a % n == 0 and b % n == 0

def frequency_groups(cells, i, j, n):
    ordered = sorted(cells, key=lambda c: hex_center(*c)[0]**2 + hex_center(*c)[1]**2)
    reps, labels = [], {}
    for q, r in ordered:
        for idx, (rq, rr) in enumerate(reps):
            if _in_cluster_lattice(q - rq, r - rr, i, j, n):
                labels[(q, r)] = idx
                break
        else:
            reps.append((q, r))
            labels[(q, r)] = len(reps) - 1
    assert len(reps) == n
    return labels
```

B.2 Shared handover decision logic (Mini Projects 3.1 and 3.3)

```
cand = max(others, key=lambda c: series[c, t])
entering = series[cand, t] > series[serving, t] + margin
streak = streak + 1 if entering else 0

if policy == "naive":
    trigger = series[cand, t] > series[serving, t]
else:
    trigger = entering and streak >= ttt

if trigger:
    handovers.append((t, serving, cand))
    serving, streak = cand, 0
```

B.3 RACH-controlled Idle → Connected transition (Mini Project 3.3)

```
log(t, "Idle", "Random-Access",
    "Connection request -> RACH preamble (Msg1)", best)
log(t + 1, "Random-Access", "Random-Access",
    "Random Access Response + timing advance (Msg2)", best)
log(t + 2, "Random-Access", "Random-Access",
    "RRC connection request (Msg3)", best)
log(t + 3, "Random-Access", "Connected",
    "Contention resolution success (Msg4)", best)
serving, state, streak = best, "Connected", 0
```

The published scripts are the authoritative version; the excerpts are shortened for readability and omit imports, figure code and the logging helpers.